Deep Learning-based Neural Granger Causality for Solar Radiation Data

Minwoo-Lee, Sangmin-Lee, Younghwi-Cho, Namyoon-Kim, Gahee-Kim, Youngmi-Lee
Department of Statistics, Jeonbuk National University

Background of the Study

One of the most widely used approaches for identifying such relationships is the Granger causality test; however, its reliance on linear assumptions limits its ability to capture the complex and nonlinear interactions frequently observed in real-world time series data. To overcome these limitations, recent studies have actively explored nonlinear extensions of Granger causality by incorporating machine learning and deep learning techniques. In particular, deep learning models such as Multi-Layer Perceptrons (MLPs) and Long Short-Term Memory (LSTM) networks have demonstrated strong predictive and classification performance on large-scale nonlinear datasets, and their integration with Granger-type inference presents a promising direction for identifying more intricate predictive structures.

In this study, we adopt the Neural Granger Causality framework proposed by Tank et al. (2021) and apply it to solar radiation in meteorological variables to investigate neural network-based Granger causal relationships, assess the effectiveness of deep learning-based causal inference, and highlight its advantages over conventional linear approaches.

Research Methodology

Statistical Granger causality

Granger causality is a statistical framework used to assess whether past values of one time series (e.g. y) provide predictive information about another time series (e.g. x) beyond what is already contained in its own past. The core idea is that causality is inferred from improved prediction: If the history of y improves the forecast of x, then, y is said to Granger – cause x. Granger argued that under certain assumptions, predictability may imply causality. This is referred to as Granger Causality. Granger causality is The definition below

$$\mathrm{Var}\left[x_t - p(x_t \mid H_{<t})\right] < \mathrm{Var}\left[x_t - p(x_t \mid H_{<t} \setminus y_{<t})\right]$$

- $H_{<t}$: All relevant information available up to time t – 1.
- $p(x_t|H_{<t})$: the optimal prediction if x given the information
- $H_{<t} \setminus y_{<t}$: the information excluding all past values of y before time t – 1
- $y_{<t}$: A different time series from x, observed up to time t – 1

This inequality states that including the past of y reduces the prediction error variance of x then, we can say Granger causes x.

Deep learning-based neural Granger causality

Vector Auto Regression (VAR) models are unable to capture nonlinear relationships and require the specification of lag orders in advance. To address these limitations, neural network-based models (MLP, LSTM) have been introduced. Although neural networks provide strong predictive performance, they present two main challenges:

- Modeling the entire time series jointly makes it difficult to isolate the effects of individual input variables on specific output variables.
- A uniform lag structure must be imposed across all variables.

$$x_{ti} = g_i(x_{<t}) + \epsilon_{ti}$$

- To overcome these limitations, Therefore, to disentangle the influence of each input variable, we build an independent model g_i for each target variable i $(i = 1, \ldots, p)$
- two models were proposed: component-wise MLP (cMLP) and component-wise LSTM (cLSTM), which employ independent neural networks for each variable along with sparsity constraints.

Model

cMLP

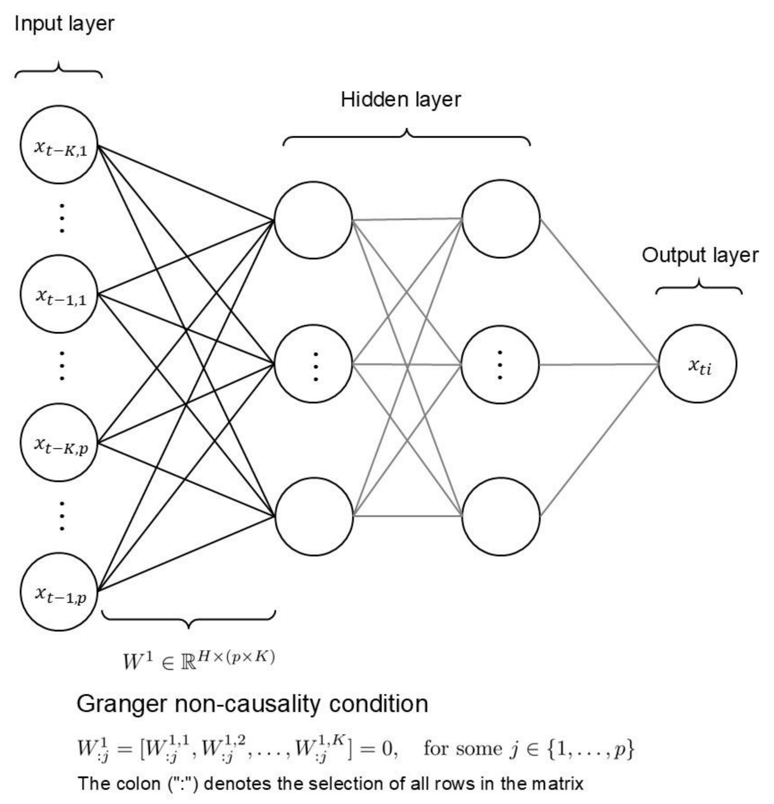

Figure 1 : Granger causality condition derived from the weights of the first layer in the cMLP

cLSTM

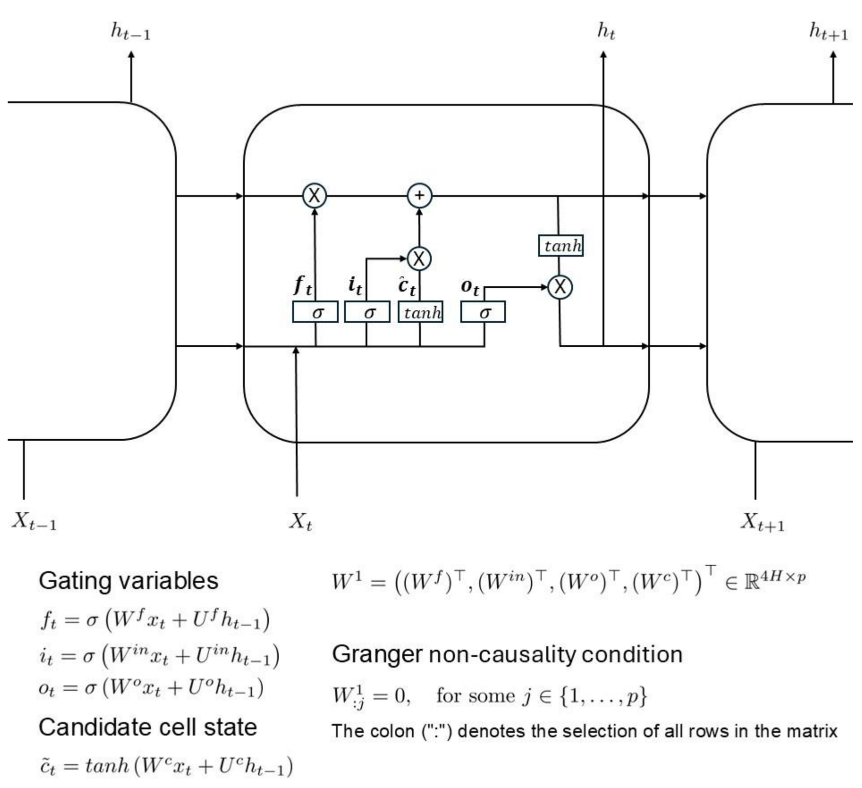

Figure 2 : Granger causality condition derived from the weights of the first layer in the cLSTM

Regularization penalty applied

Sparse Group Lasso

$$\alpha \left\|W^1_j\right\|_F + (1-\alpha)\sum_{k=1}^{K} \left\|W^{1,k}_j\right\|_2$$

$\alpha \in (0,1)$, $\|\cdot\|_F$: Frobenius Norm

- This penalty assumes that only a few lags of a series j are predictive of series i
- Three types of penalization are considered for cMLP : group lasso, sparse group lasso and group sparse group lasso.
- We adopt sparse group lasso because it provides more stable performance

Group Lasso

$$\min_{\mathbf{W}} \sum_{t=2}^{T} \left(x_{it} - g_i(x_{<t})\right)^2 + \lambda \sum_{j=1}^{p} \left\|W^1_{:j}\right\|_2$$

- For a large enough λ, many columns of W^1 will be zero , leading to a sparse set of Granger causal connections
- The cLSTM model requires no maximum lag specification, and instead automatically learns the memory of each interaction.

Data Description

Meteorological Variables based on hourly observed data in Jeon-ju from 2023/01/01 to 2023/12/31.

RAIN: Precipitation (mm)
TEM: Temperature (°C)
HUM: Humidity (%)
SOL: Solar Radiation (MJ/m²)
WIND: Wind Speed (m/s)
CLOUD: Total Cloud Cover (in tenths)

Table1 : Summary statistics for weather variables.

	RAIN	TEM	HUM	SOL	WIND
count	8760	8760	8760	8760	8760
mean	0.1550	15.7993	71.0723	0.5856	1.5505
std	1.1694	10.1662	17.0081	0.8879	1.0017
min	0.0000	-8.6000	14.0000	0.0000	0.0000
25%	0.0000	6.6000	59.0000	0.0000	0.9000
50%	0.0000	16.2000	73.0000	0.0200	1.3000
75%	0.0000	25.0000	85.0000	0.9600	2.0000
max	29.6000	36.2000	99.0000	3.5300	7.8000

Table2 : Frequency of Cloud Cover

CLOUD	Frequency
0	2323
1	237
2	346
3	358
4	343
5	407
6	615
7	611
8	730
9	813
10	1977

Research Results

Experimental algorithm

Algorithm 1: Deep learning-based training algorithm

Input: Data set : $D = \{(X^{(i)}, Y^{(i)})\}_{i=1}^{T}$, Lag : K
Output: Inference result
for $i \leftarrow 1$ **to** 50 **do**
  Sample index $t \sim$ Bootstrap from $\{K+1, \ldots, T\}$
  Construct training sample $D' = (X_{t-K}, \ldots, X_{t-1};\ Y_t)$
  Train model $\mathcal{M}$ using D'
  Perform granger calusality inference of Y_t in $\mathcal{M}$
return Inferce result

- This is the experimental algorithm for the neural-based model.
- The data has been normalized using min-max scaling.
- The input and output dimensions of D' are as follows: $X \in \mathbb{R}^{K\times p}, Y \in \mathbb{R}$.
- This provides inference results for the Y_t.
- If Granger causality is present, the result is 1; otherwise, it returns 0.
- The VAR model conducted the Granger causality test with a significance level of 0.005.

Result

Table 3. Granger causality results of predictors influencing SOL.

Data	Model		
	VAR	cMLP	LSTM
TEM	1.00	1.00	1.00
RAIN	0.37	0.79	0.97
WIND	0.74	0.72	0.97
HUM	0.74	0.93	1.00
CLOUD	0.60	0.65	0.79

- This is the result table of the Granger causality test for SOL when K = 12.
- The test was repeated 50 times, and the results were aggregated for each variable. The variables were ranked based on the most frequently identified causal relationships.
- In all three models, TEM was the most frequently identified variable, followed by HUM as the second most frequent.
- RAIN and CLOUD were more frequently detected by the deep learning-based models compared to the VAR model.

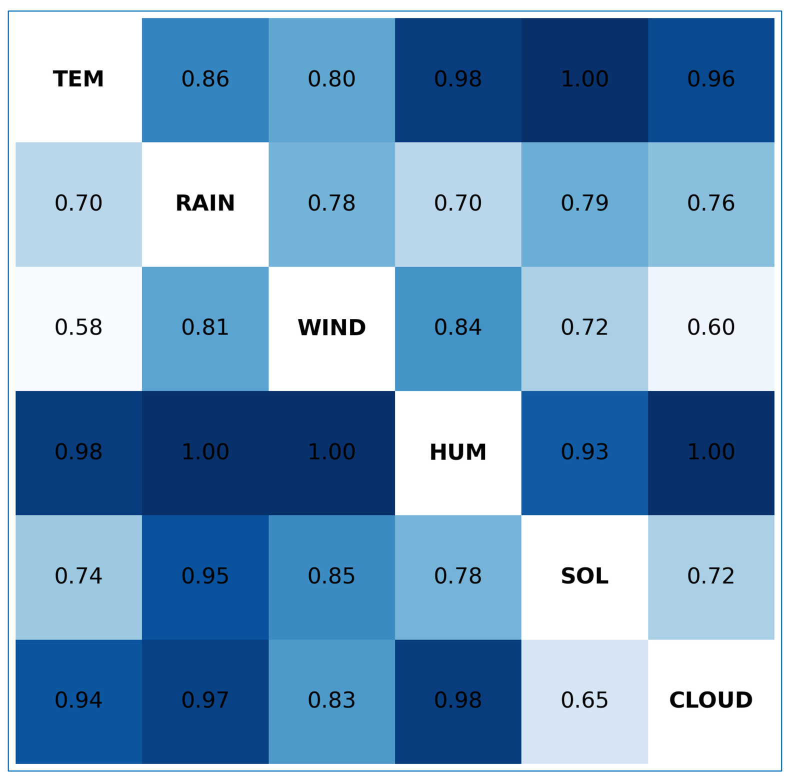

Figure 3 : Granger causality for each meteorological variables using the cMLP

Figure 4 : Granger causality for each meteorological variables using the cLSTM

Neural Granger causality was applied to six meteorological variables, and the results were visualized as a heatmap. The vertical axis indicates the variables that are potential Granger causes, while the horizontal axis represents the variables that are potentially affected.

Conclusion

Neural Granger causality analysis was performed using meteorological variables obtained from the Korea Automated Synoptic Observing System. As shown in Table 3, for the target variable SOL, the models produced consistent results for continuous variables such as TEM and HUM, while notable discrepancies were observed for discrete variables like RAIN and CLOUD. This suggests that deep learning models may be better suited to capturing nonlinear relationships in continuous data. However, there is a lack of prior research that provides theoretical justification for the application of deep learning-based models in this context. Moreover, the experiment was conducted using only a single lag (K = 12), and due to the high sensitivity of these models to hyperparameters, it is difficult to assert the robustness of the results.

References and Resources

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